

Identify Business Entities

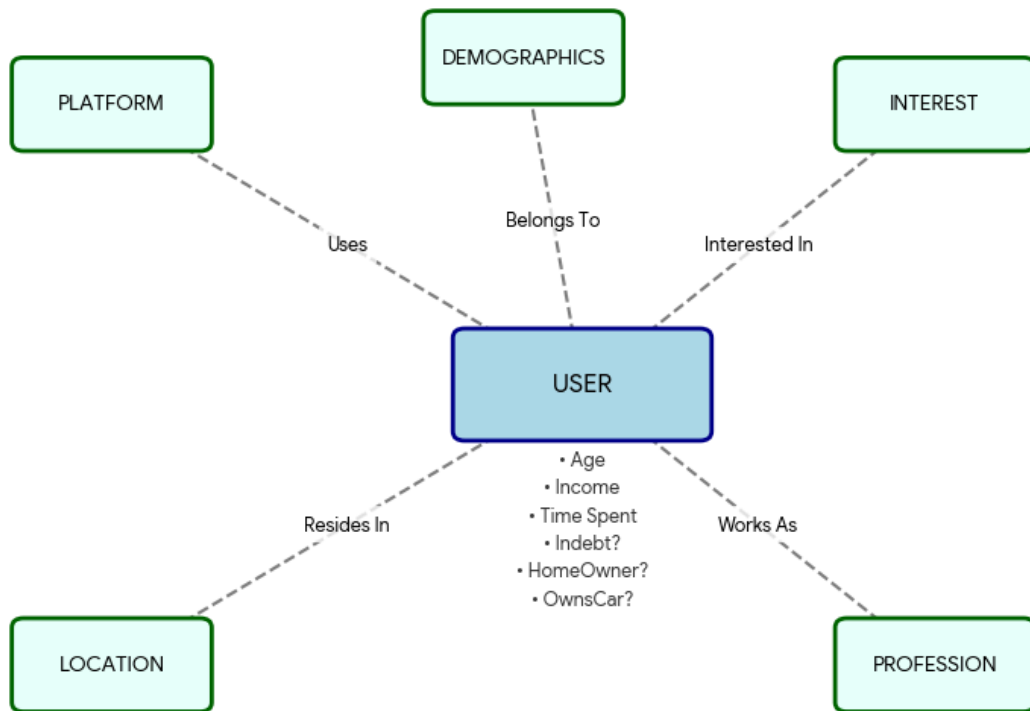
- age, gender, demographics → **Customer**
- platform, time_spent → **Platform Usage**
- interests → **Interest**
- location → **Location**
- profession, income → **Profession / Financial Profile**
- indebt, isHomeOwner, Owns_Car → **Financial Status**

Final Entities:

1. **Customer**
2. **Platform**
3. **Interest**
4. **Location**
5. **Profession**
6. **Financial Status**

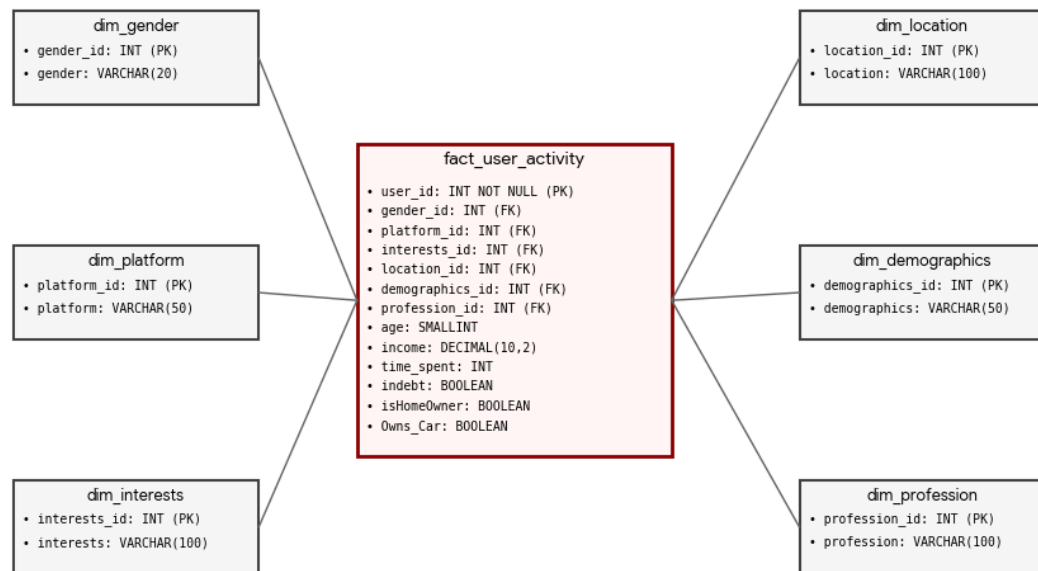
Design conceptual data model.

Conceptual Data Model: User Activity Analysis



logical and physical data model:

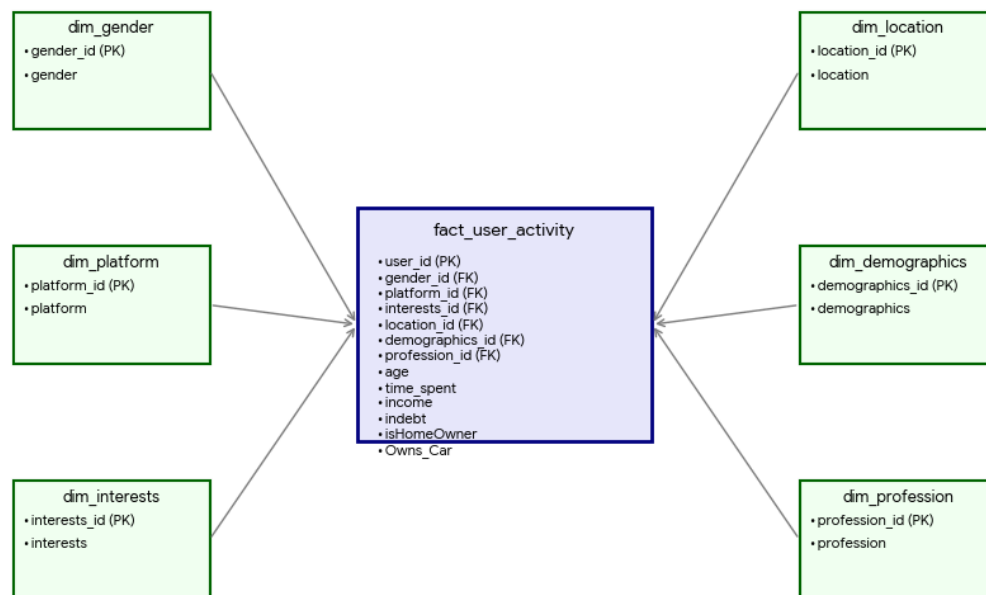
Physical Data Model



STAR SCHEMA:

:

Star Schema for Social Media User Activity



5. Define Fact and Dimension Tables

In this schema, we separate the "numbers" (Facts) from the "context" (Dimensions).

Fact Table: `fact_user_activity`

This table records the specific quantitative metrics for every user.

- **Foreign Keys (FK):** `gender_id`, `platform_id`, `interests_id`, `location_id`, `demographics_id`, `profession_id`.
- **Measures (Facts):** * `time_spent`: Amount of daily hours on a platform.
 - `income`: Annual income of the user.
 - `age`: Physical age of the user.
- **Degenerate Dimensions/Flags:** `indebt`, `isHomeOwner`, `Owns_Car` (these can be treated as facts to calculate "counts" or "percentages").

Dimension Tables:

These provide the "lookup" values to filter the facts.

- **`dim_platform`:** Contains `platform_name` (Facebook, Instagram, YouTube).
- **`dim_location`:** Contains geographic data (United Kingdom, Australia, United States).
- **`dim_profession`:** Contains job titles (Software Engineer, Student, Marketer Manager).
- **`dim_interests`:** Contains user hobbies (Sports, Travel, Lifestyle).
- **`dim_demographics`:** Contains environmental types (Urban, Sub_Urban, Rural).

6. Identify KPIs and Business Rules

These are the goals and constraints your data model must support.

Key Performance Indicators (KPIs)

1. **Avg. Engagement Rate:** `AVERAGE(time_spent)` grouped by `platform`.
2. **Affluence Score:** `AVERAGE(income)` grouped by `profession` or `location`.
3. **Debt-to-Ownership Ratio:** Percentage of users where `indebt = True` compared to `isHomeOwner = True`.
4. **Platform Dominance:** Count of users per `platform` within specific `demographics`.
5. **Interest-Income Correlation:** Comparison of `income` across different `interests` (e.g., do "Travel" enthusiasts earn more than "Sports" fans?).

Business Rules

- **Non-Negative Values:** `time_spent`, `income`, and `age` must always be greater than or equal to zero.
- **Platform Specificity:** A single user record in the fact table must link to exactly one primary platform.
- **Boolean Consistency:** Ownership flags (`Owns_Car`, etc.) must be strictly Boolean (True/False).
- **Income Bracketing:** Any income over \$200,000 should be flagged for verification (Outlier Detection).

7. Data Granularity and Hierarchies

Understanding the "level" of your data ensures accurate reporting.

Data Granularity

The granularity of this dataset is **Atomic Individual Level**.

- **One row = One unique user profile snapshot.** * You cannot break this down further (e.g., you don't have "time spent per day" over a month, just a single average per user).

Hierarchies

Hierarchies allow you to "drill down" from high-level summaries to specific details.

1. **Geography Hierarchy:**
 - Global $\rightarrow$ Country (Location) $\rightarrow$ Environment Type (Demographics).
2. **Economic Hierarchy:**
 - Income Bracket (e.g., High Earner) $\rightarrow$ Debt Status $\rightarrow$ Asset Ownership (Home/Car).
3. **Life Stage Hierarchy:**
 - Age Group (e.g., Gen Z, Millennial) $\rightarrow$ Profession (e.g., Student, Professional).